



Dynamic Bond ETF Allocation Using Nelson-Siegel Yield Curve Mispricing Signals

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Abstract

This paper develops a systematic fixed income allocation strategy across U.S. Treasury ETFs by combining the Nelson-Siegel yield curve model with the Diebold-Li dynamic factor extension. Each month, Nelson-Siegel is fit to the Treasury par yield curve, extracting three factors, level (β_0), slope (β_1), and curvature (β_2), along with cross-sectional mispricing residuals at each ETF's effective maturity. Following Diebold and Li (2006), the factors are modeled jointly as a VAR(1) system to generate one-month-ahead yield curve forecasts on an expanding window. Portfolio construction is driven by two signals: a mispricing signal derived from Nelson-Siegel residuals and a directional signal from the VAR(1) factor forecasts, with factor loadings tilted 60/20/20 toward slope based on individual factor attribution. Signal-proportional and top-N concentration weighting schemes are backtested against an equal-weight benchmark across seven Treasury ETFs from 2008 to 2024, with Sharpe ratio significance assessed via the Jobson-Korkie (1981) test. A sub-period analysis over 2018–2021 stress-tests the model through the 2019 Federal Reserve rate-cut cycle and the 2020 COVID-19 shock. Results indicate that the slope factor β_1 carries the dominant predictive power, and that layering curve mispricing onto directional forecasts produces consistent risk-adjusted outperformance relative to passive equal-weighting.



1. Introduction

The yield curve is arguably the single most information-rich signal in financial markets. It simultaneously reflects market expectations for monetary policy, inflation, and economic growth, encoded in a single object observable in real time. For a portfolio manager allocating across Treasury ETFs, forecasting where the curve moves next and identifying maturities that are temporarily mispriced is the direct mechanism for generating alpha over a passive benchmark. The fixed-income ETF market has grown substantially over the past two decades, and that expansion has made duration-targeting strategies practical. The tight spreads, deep liquidity, and daily price transparency make the Treasury ETF universe one of the cleanest testing environments available for a yield-curve-based allocation model.

The Nelson-Siegel (1987) model fits the entire yield curve using just three parameters. A level factor controls the overall height of the curve, a slope factor governs the steepness from short to long maturities, and a curvature factor captures the hump or trough at intermediate maturities. The model has become a standard tool across central banks and fixed-income practitioners for good reason. It is consistent with the well-established empirical finding that three latent factors explain the vast majority of yield curve variation (Litterman and Scheinkman, 1991). The Bank for International Settlements (2005) documents its adoption across numerous central banks for official yield curve estimation, supporting both its practical reliability and its theoretical grounding.

Diebold and Li (2006) extended the Nelson-Siegel framework by treating the three parameters as time-varying factors and modeling their joint dynamics with a VAR. Fixing the decay parameter and casting the cross-sectional regression as the measurement equation of a state-space system, they showed that the dynamic Nelson-Siegel model produces out-of-sample yield forecasts that are competitive with, and often better than, random walk, slope regression, and affine term structure benchmarks, especially when examining horizons of six months and beyond.

This project applies the Nelson-Siegel and Diebold-Li framework to dynamic allocation across U.S. Treasury ETFs. The economic intuition behind the strategy is straightforward. When the yield at a given maturity deviates from the Nelson-Siegel fitted value, that gap represents a potential mispricing along the term structure. One that should revert as the curve reasserts its fair-value shape. When the VAR(1) forecasts a decline in yields at a particular maturity, ETFs with duration exposure at that point on the curve should see capital gains and outperform. With two signals one cross-sectional relative value signal and one time-series directional signal we can combine them to produce a composite allocation that captures both mean reversion in relative yields and directional momentum in the level and shape of the curve.

The methodology proceeds in four stages. Nelson-Siegel factors are estimated each month via cross-sectional OLS with a fixed decay parameter, following the two-step procedure in Diebold and Li (2006) and Diebold et al. (2008). Multicollinearity between the slope and curvature regressors is verified using the condition number diagnostic from Annaert et al. (2010), which confirms



that OLS is appropriate for the maturity vector used here without requiring ridge regression. An expanding-window VAR(1) then models the joint dynamics of the three factors, generating one-month-ahead forecasts that respect the information available at each rebalancing date. Individual factor attribution isolates the contribution of each Nelson-Siegel factor to cross-ETF return predictability. The slope factor emerges as the dominant source of predictive power, motivating a 60/20/20 factor-weighted tilt in the directional signal. The combined signal is then translated into portfolio weights and backtested against an equal-weight benchmark across seven Treasury ETFs spanning effective durations from under one year to approximately 25 years.

Our main finding is that the signal-proportional strategy outperforms the equal-weight benchmark on a risk-adjusted basis over the full 2008–2024 sample, as measured by the Sharpe ratio. The slope factor is the dominant contributor to performance throughout, consistent with the established literature linking yield curve slope to business cycle dynamics, monetary policy expectations, and future bond returns. The Diebold-Li directional signal outperforms the Nelson-Siegel mispricing signal in isolation — suggesting the time-series forecasting component drives more of the alpha than the cross-sectional relative value component. Although combining the two signals beats either one alone. Sub-period analysis over 2018–2021 confirms the strategy adds the most value when the yield curve is moving directionally. The 2019 Fed rate-cut cycle and the 2020 COVID-19 shock are the clearest examples of where the VAR(1) successfully anticipated the direction of curve shifts before they fully played out.

The rest of the paper is structured as follows. Section 2 surveys the relevant literature on yield curve modeling, dynamic Nelson-Siegel estimation, multicollinearity diagnostics, and fixed-income portfolio strategies. Section 3 covers the data sources, descriptive statistics, and full methodology — including the Nelson-Siegel OLS estimation, Diebold-Li VAR(1) dynamics, signal construction, and backtesting procedure. Section 4 presents the empirical results: signal decomposition, factor attribution, full-period and sub-period performance, and significance testing. Section 5 concludes with a summary of the main findings, a discussion of limitations, and directions for future research.

2. Literature Survey

The yield curve modeling literature spans several decades and encompasses parametric, spline-based, and latent-factor approaches. Early contributions by McCulloch (1971, 1975) used cubic splines to interpolate yields across maturities, while Vasicek and Fong (1982) proposed exponential splines. These methods, while flexible, were criticized for undesirable economic properties such as oscillatory behavior at long maturities and for being opaque in their economic interpretation. The need for interpretable alternative motivated the parametric approach of Nelson and Siegel (1987).

Nelson and Siegel (1987) proposed a three-parameter model for the forward rate curve that decomposes into a constant (level), an exponential decay (slope), and a Laguerre function (curvature). The corresponding spot rate function, obtained by averaging the forward curve, provides a flexible representation of the yield curve that can accommodate upward-sloping, downward-sloping,



humped, and S-shaped term structures. The model's appeal lies in its factor interpretation, which aligns with the empirical finding of Litterman and Scheinkman (1991) that three principal components: level, slope, and curvature explain over 99 percent of the variation in U.S. Treasury bond returns. Svensson (1994) extended the Nelson-Siegel model with a second curvature term, allowing for more complex shapes such as double humps, though the additional parameter introduces estimation challenges.

The estimation of the Nelson-Siegel model presents significant challenges. The model is nonlinear in the decay parameter λ , and nonlinear optimization is sensitive to starting values, as demonstrated by Cairns and Pritchard (2001). To circumvent this problem, Nelson and Siegel (1987) originally proposed fixing λ on a grid and selecting the value that maximizes R^2 . Diebold and Li (2006) simplified further by fixing λ at a single value (0.0609 monthly, corresponding to approximately 1.37 annualized) that places the curvature loading maximum at 30 months. This linearizes the model and reduces estimation to OLS, a key practical advantage.

Annaert et al. (2010) provided the most systematic analysis of the multicollinearity problem inherent in Nelson-Siegel estimation. They demonstrated that the correlation between the slope and curvature regressors depends critically on the fixed decay parameter and the vector of maturities used in estimation. Using four different maturity vectors and two common choices of λ (1.37 and 3), they showed that regressor correlations can range from 0.256 to -0.931 depending on the configuration. They proposed measuring the degree of multicollinearity using the condition number of the regressor matrix, with a threshold of 10 following Belsley (1991), and applying ridge regression whenever the threshold is exceeded. Their grid search with conditional ridge regression produced the lowest out-of-sample mean absolute prediction errors across all maturities, including during the 2008–2009 financial crisis. For our study, their condition number diagnostic provides a formal validation that OLS estimation is appropriate for our maturity vector and choice of λ , without the need for ridge regression.

Diebold and Li (2006) represent the foundational contribution on dynamic Nelson-Siegel modeling. By treating the three Nelson-Siegel parameters as time-varying latent factors and fitting them month by month, they obtained factor time series that they then modeled with autoregressive processes. They showed that a simple AR(1) model for each factor, or equivalently a diagonal VAR(1), produces yield forecasts that outperform random walk, Fama-Bliss, and other standard benchmarks at horizons of 6 to 12 months. The factors have clear economic interpretations: the level factor correlates with long-run inflation expectations, the slope factor tracks the business cycle and the stance of monetary policy, and the curvature factor captures medium-term dynamics in the hump of the yield curve.

Diebold et al. (2008) extended this framework to a multi-country setting, constructing a hierarchical dynamic factor model in which country yield factors load on global yield factors as well as country-specific components. Using government bond yields for the United States, Germany, Japan, and the United Kingdom from 1985 to 2005, they estimated the model using Bayesian



MCMC methods with Gibbs sampling and the Carter-Kohn (1994) algorithm for factor extraction. Their key findings were that a global level factor explains 60 to 80 percent of the variation in country-level factors, that the global level factor is linked to global inflation trends, and that the global slope factor tracks the international business cycle. The importance of global factors was found to increase in the second half of their sample, consistent with growing international bond market integration. Their two-step estimation procedure, first extracting factors via cross-sectional OLS, then modeling dynamics provides the methodological template for our study, adapted to a single-country context.

The connection between yield curve factors and macroeconomic variables has been explored extensively. Ang and Piazzesi (2003) developed a no-arbitrage VAR with both latent yield factors and observable macro factors, finding that inflation and real activity significantly improve yield curve forecasts. Diebold et al. (2006) embedded the dynamic Nelson-Siegel model in a macro-finance framework, documenting bidirectional interactions between the yield curve and the macroeconomy: the level factor responds to inflation, the slope factor responds to real activity, and yield curve factors in turn Granger-cause macroeconomic variables. These findings provide the economic rationale for the predictive power of the slope factor in our allocation strategy.

The application of yield curve models to active bond portfolio management has received comparatively less attention in the academic literature. Barrett et al. (1995) used Nelson-Siegel factors for yield curve immunization strategies. Fabozzi et al. (2005) demonstrated the predictive content of the Nelson-Siegel shape factors for future yield curve movements, benchmarking forecasts against alternative models and finding that Nelson-Siegel performs well especially for longer forecast horizons. Martellini and Meyfredi (2007) used the Nelson-Siegel approach for value-at-risk estimation of fixed-income portfolios.

Our study contributes to this strand of the literature by developing a specific, implementable trading strategy that combines Nelson-Siegel mispricing signals with Diebold-Li directional forecasts, backtested on a universe of liquid Treasury ETFs. The main methodological contributions relative to the existing literature are the dual-signal framework that combines cross-sectional and time-series information, the systematic factor attribution that motivates the slope-tilted weighting scheme, and the use of the Annaert et al. (2010) condition number diagnostic to formally validate the estimation procedure.

3. Data and Methodology

3.1 Data Sources and Construction

Our dataset consists of two components: Treasury par yield curves and bond ETF prices. Treasury par yields are obtained from the Federal Reserve Economic Data (FRED) database, published by the Federal Reserve Bank of St. Louis, for seven constant maturities: 2, 3, 5, 7, 10, 20, and 30 years. The FRED series identifiers are DGS2, DGS3, DGS5, DGS7, DGS10, DGS20, and



DGS30, respectively. The raw data are daily observations spanning January 2008 to December 2024. Missing values, which FRED codes as periods (.) on days when Treasury auctions did not settle, are treated as missing and excluded. Daily yields are resampled to month-end frequency by taking the last available observation in each calendar month, and are converted from percentage to decimal form for estimation.

Bond ETF prices are obtained via the Yahoo Finance API for seven iShares and Vanguard Treasury ETFs that collectively span the duration spectrum: SHV (iShares Short Treasury Bond ETF, effective duration approximately 0.5 years), SHY (iShares 1–3 Year Treasury Bond ETF, approximately 2 years), IEI (iShares 3–7 Year Treasury Bond ETF, approximately 4.5 years), IEF (iShares 7–10 Year Treasury Bond ETF, approximately 8.5 years), TLH (iShares 10–20 Year Treasury Bond ETF, approximately 15 years), TLT (iShares 20+ Year Treasury Bond ETF, approximately 20 years), and EDV (Vanguard Extended Duration Treasury ETF, approximately 25 years). Adjusted close prices, which account for distributions, are resampled to month-end frequency. ETFs with less than 80 percent non-missing observations over the sample period are excluded to ensure adequate coverage; EDV, which launched in December 2007, survives this filter. The Treasury yield and ETF price series are aligned on a common set of month-end dates, yielding approximately 200 aligned monthly observations.

We deliberately exclude Treasury Inflation-Protected Securities (TIPS) ETFs such as TIP from our universe. Our entire strategy maps ETFs onto a nominal Treasury yield curve via the Nelson-Siegel model. An inflation-linked instrument would produce fitting residuals driven primarily by the breakeven inflation premium rather than by exploitable mispricings along the nominal term structure. Including TIPS would introduce a systematic bias in both the mispricing and directional signals, as the model would interpret the inflation premium as a yield deviation from the nominal curve.

3.2 Descriptive Analysis

Treasury Yield Summary Statistics

Table 1 presents summary statistics for monthly Treasury par yields across the seven maturities over the 2008–2024 sample period. The average yield curve is upward-sloping, with the 2-year yield averaging substantially below the 30-year yield, reflecting the typical term premium structure. Yield volatility is highest at the short end during this sample, driven by the Federal Reserve’s dramatic policy shifts between near-zero rates (2009–2015 and 2020–2022) and the rapid tightening cycle of 2022–2023. All yield series exhibit positive skewness, reflecting the lower bound on nominal rates, and mild excess kurtosis consistent with fat-tailed behavior. The 10-year minus 2-year term spread provides a particularly informative summary of yield curve dynamics: it was persistently positive through most of the sample but inverted notably in the 2022–2023 period, reflecting market expectations of a recession in the wake of aggressive monetary tightening.



Table 1: Treasury Yield Summary Statistics (% , 2008–2024)

Maturity	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
2Y	1.47	1.43	0.11	5.07	1.20	0.26
3Y	1.63	1.30	0.11	4.90	1.06	0.05
5Y	1.99	1.10	0.21	4.82	0.70	-0.23
7Y	2.32	1.01	0.39	4.89	0.41	-0.42
10Y	2.60	0.95	0.55	4.88	0.15	-0.58
20Y	3.11	0.97	0.98	5.21	0.07	-0.88
30Y	3.24	0.88	1.20	5.04	-0.11	-0.72

ETF Return Summary Statistics

Table 2 reports annualized return and risk statistics for the seven Treasury ETFs. As expected, longer-duration ETFs exhibit higher annualized volatility and larger maximum drawdowns, consistent with the mechanical relationship between duration and interest rate sensitivity. SHV, with an effective duration of approximately 0.5 years, serves as a near-cash instrument with minimal volatility, while EDV, with an effective duration of approximately 25 years, exhibits the most extreme risk-return profile. The Sharpe ratios vary across the duration spectrum and across sub-periods, with intermediate-duration ETFs (IEI, IEF) generally providing the best risk-adjusted returns over the full sample. Return distributions for all ETFs exhibit mild negative skewness and positive excess kurtosis relative to the normal distribution, as confirmed by histograms overlaid with normal density fits. These departures from normality are consistent with the well-documented fat-tailed behavior of fixed-income returns and motivate the use of the Sortino ratio alongside the Sharpe ratio as a risk-adjusted performance measure.

Table 2: ETF Monthly Return Summary Statistics (2008–2024)

ETF	Ann. Return(%)	Ann. Vol(%)	Sharpe	Skewness	Kurtosis	Max DD(%)
SHV	5.03	21.94	0.23	0.60	1.87	-59.15
SHY	2.74	6.85	0.40	0.22	0.72	-23.15
IEI	2.20	3.93	0.56	-0.50	0.57	-13.59
IEF	1.13	0.51	2.22	1.38	0.79	-0.30
TLH	1.32	1.43	0.93	0.23	2.12	-5.36
TLT	2.78	10.38	0.27	0.32	0.99	-40.23
EDV	3.53	14.66	0.24	0.46	1.09	-47.61

Correlation Structure

The cross-sectional correlation matrix of monthly ETF returns reveals the expected monotonic pattern as shown in Figure 1, ETFs adjacent in the duration spectrum are highly correlated (SHY-IEI, IEF-TLH), while the correlation between the shortest-duration (SHV) and longest-duration



(EDV) ETFs is substantially lower. This correlation structure is important for the strategy because it determines the diversification benefits of reallocating across the duration spectrum; if all ETFs were perfectly correlated, there would be no value in dynamic allocation. We also compute the correlation matrix of monthly yield changes across the seven Treasury maturities. The high cross-maturity correlation in yield changes (all pairwise correlations above 0.5) provides the empirical motivation for using a low-dimensional factor model such as Nelson-Siegel: if yield changes were independent across maturities, a three-factor model would have little descriptive power.

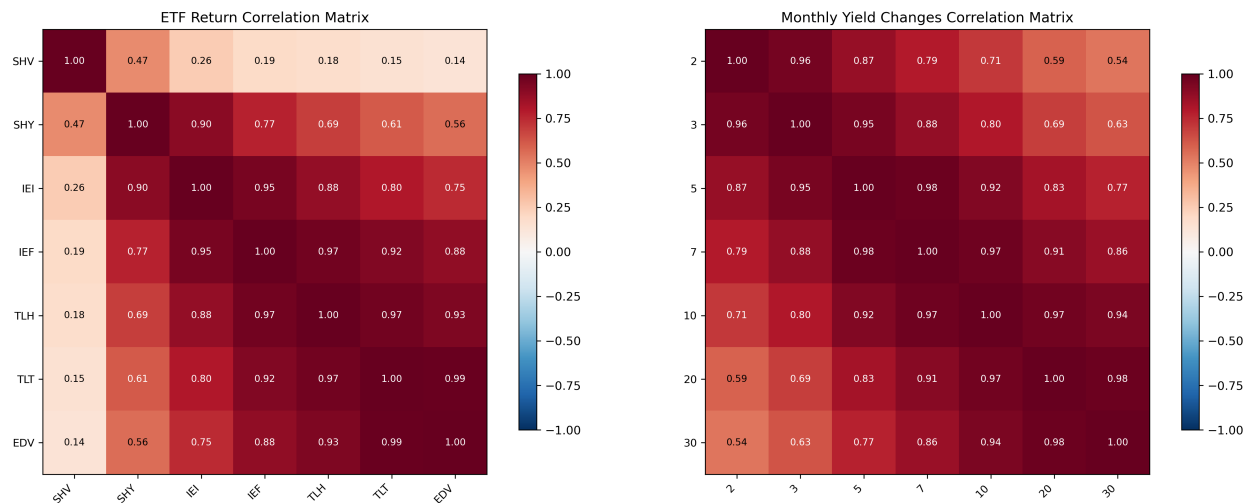


Figure 1: ETF Return and Monthly Yield Changes Correlation Matrices

Time-Series Dynamics

The time-series plots, as shown in Figure 2, of Treasury par yields at selected maturities (2, 5, 10, and 30 years) document the major yield curve regimes over the sample period: the post-crisis decline to near-zero short rates (2008–2015), the gradual normalization (2016–2019), the COVID-19 shock and return to zero rates (2020–2021), the aggressive tightening cycle (2022–2023), and the beginning of the easing cycle in 2024. The 10-year minus 2-year spread panel highlights the three inversions during the sample. Twelve-month rolling annualized volatility plots for selected ETFs (SHY, IEF, TLT, EDV) reveal pronounced time variation in risk, with volatility spikes during the 2008 financial crisis, the 2013 taper tantrum, the 2020 pandemic crisis, and the 2022 rate shock. This time-varying risk environment is precisely the setting where a dynamic allocation model can add the most value.

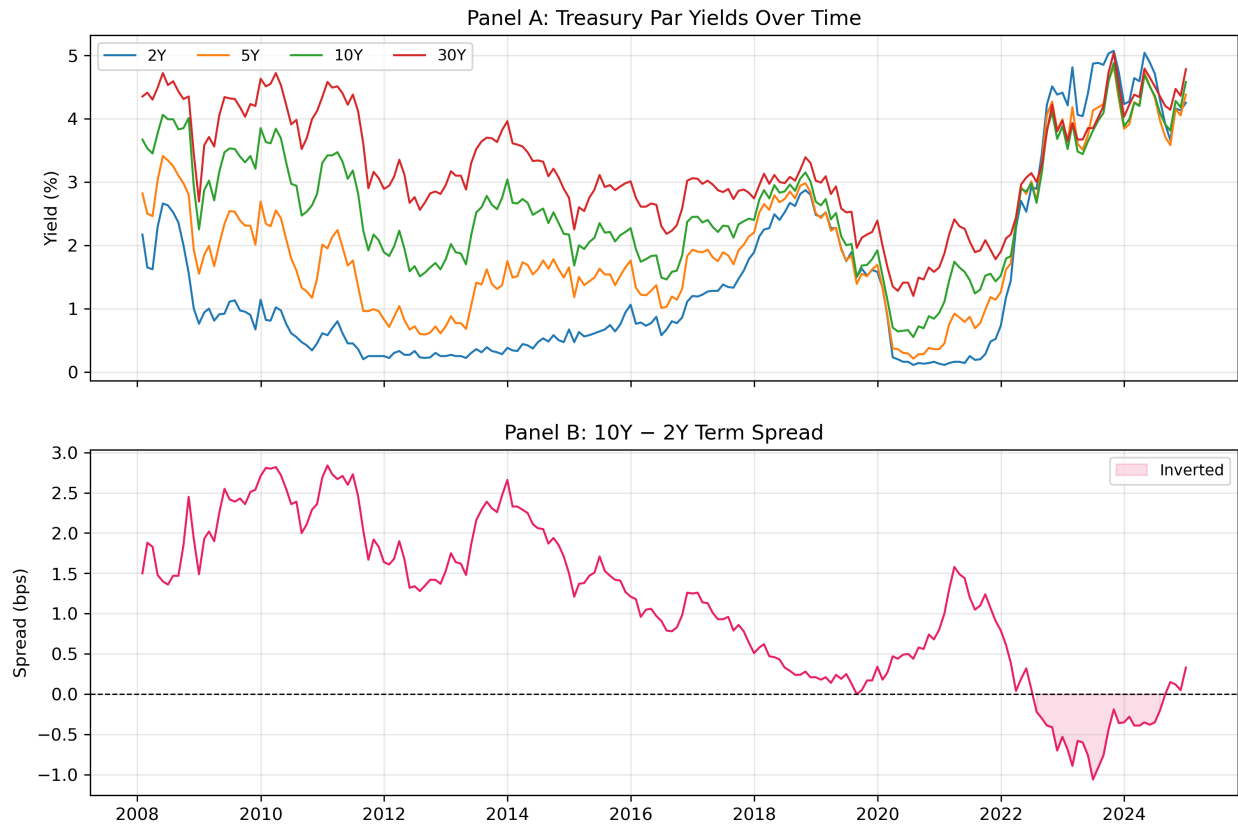


Figure 2: Treasury Par Yields and 10Y - 2Y spread

3.3 Model Estimation

3.3.1 Nelson-Siegel Curve Fitting

The Nelson-Siegel (1987) model, given in Equation (1), expresses the continuously compounded zero-coupon yield at maturity τ as:

$$y(\tau) = \beta_0 + \beta_1 \cdot \frac{1 - e^{-\lambda\tau}}{\lambda\tau} + \beta_2 \cdot \left[\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right] \quad (1)$$

where β_0 is the level factor, β_1 is the slope factor, β_2 is the curvature factor, and $\lambda > 0$ is the decay parameter. The first loading is a constant (unity), identifying β_0 as the long-run yield level. The second loading starts at one for short maturities and decays to zero, so β_1 captures the spread between short and long rates. The third loading starts at zero, rises to a maximum at an intermediate maturity, and then decays back to zero, so β_2 governs the hump or trough in the middle of the curve. The decay parameter λ controls the rate of exponential decay and determines the maturity at which the curvature loading reaches its peak.

Following Diebold and Li (2006), we fix $\lambda = 0.0609$ (monthly), which places the maximum of



the curvature loading at approximately 30 months (≈ 2.5 years). With λ fixed, the model becomes linear in $[\beta_0, \beta_1, \beta_2]$, and estimation reduces to a cross-sectional OLS regression for each month t , as shown in Equation (2):

$$\mathbf{y}_t = \mathbf{X}\boldsymbol{\beta}_t + \boldsymbol{\varepsilon}_t \quad (2)$$

where $\mathbf{y}_t$ is the 7×1 vector of observed Treasury yields, $\mathbf{X}$ is the 7×3 matrix of Nelson-Siegel loadings (computed once from the maturity vector and λ), and $\boldsymbol{\beta}_t = [\beta_{0t}, \beta_{1t}, \beta_{2t}]'$ is obtained as the standard OLS estimator. The residual vector $\boldsymbol{\varepsilon}_t$ captures the deviation of observed yields from the Nelson-Siegel fitted curve at each maturity, and serves as the basis for the mispricing signal.

Before proceeding with OLS, we verify the absence of problematic multicollinearity using the condition number diagnostic of Annaert et al. (2010). The condition number κ is computed as the ratio of the maximum to the minimum singular value of the slope-curvature regressor submatrix $\mathbf{X}[:, 1 : 3]$. Annaert et al. (2010) establish that $\kappa < 10$ indicates that OLS produces stable estimates, while $\kappa \geq 10$ warrants ridge regression. For our configuration (maturities 2, 3, 5, 7, 10, 20, 30 years; $\lambda = 0.0609$), the condition number falls well below 10, confirming that the Diebold-Li parametrization avoids the collinearity trap documented by Annaert et al. (2010) for other maturity- λ configurations.

Unlike Diebold et al. (2008), who drop the curvature factor in their multi-country analysis due to estimation precision concerns with sparse maturity data, we retain β_2 . Our seven-maturity U.S. Treasury data from 2 to 30 years provides sufficient cross-sectional variation to identify the curvature factor reliably. The factor attribution analysis in Section 4 confirms that curvature contributes meaningfully to the allocation signal.

3.3.2 Diebold-Li VAR(1) Dynamics

Following the second step of the Diebold-Li two-step procedure, we model the joint dynamics of the extracted factor vector $\mathbf{F}_t = [\beta_{0t}, \beta_{1t}, \beta_{2t}]'$ with a VAR(1), given in Equation (3):

$$\mathbf{F}_t = \mathbf{c} + \boldsymbol{\Phi} \mathbf{F}_{t-1} + \mathbf{u}_t \quad (3)$$

where $\mathbf{c}$ is a 3×1 vector of intercepts, $\boldsymbol{\Phi}$ is the 3×3 coefficient matrix, and $\mathbf{u}_t$ is the innovation vector. Diebold et al. (2008) restrict $\boldsymbol{\Phi}$ to be diagonal in their multi-country setting, citing little cross-factor interaction in single-country analyses and the need for tractability in their Bayesian MCMC estimation. In our single-country framework, we estimate the full $\boldsymbol{\Phi}$ matrix, which allows for cross-factor dynamics such as the lagged effect of slope changes on subsequent curvature movements.

The VAR(1) is estimated using an expanding window with a 36-month burn-in period. At each month t (for $t = 37, 38, \dots, T$), the VAR is fit on all available factor observations from $t = 1$



through t , and a one-step ahead forecast is generated in Equation (4):

$$\hat{\mathbf{F}}_{t+1} = \hat{\mathbf{c}}_t + \hat{\mathbf{\Phi}}_t \mathbf{F}_t \quad (4)$$

where the hat notation indicates estimates based on the expanding window ending at t . This procedure ensures that the forecast at each point in time uses only information available to the investor at the rebalancing date, strictly preventing look-ahead bias. The 36-month burn-in is necessary to ensure a sufficient number of observations for stable VAR estimation in the initial periods.

3.4 Portfolio Trading Strategy

3.4.1 Signal Construction

The allocation strategy combines two complementary signals, each exploiting a different dimension of information in the yield curve.

The *mispricing signal* is constructed from the Nelson-Siegel residuals. For each ETF with effective maturity m , the residual at maturity m is interpolated from the monthly residual curve using linear interpolation across the seven Treasury maturities. A positive residual indicates that the observed yield at maturity m exceeds the Nelson-Siegel fitted value, suggesting that bonds at this maturity point are relatively cheap. A negative residual indicates richness. The cross-section of interpolated residuals is z-scored each month to produce a standardized mispricing signal.

The *directional signal* is constructed from the VAR(1) factor forecasts. For each ETF with effective maturity m , the implied yield change is computed as the difference between the forecast yield (reconstructed from the forecast factors) and the current yield (reconstructed from the current factors) at maturity m . The signal is negated so that a forecast yield decline produces a positive allocation signal, since declining yields correspond to rising bond prices. Within the directional signal, the forecast yield change at each maturity is decomposed into contributions from the three factors, weighted 60 percent to slope (β_1), 20 percent to level (β_0), and 20 percent to curvature (β_2). This tilt is motivated by the factor attribution results showing that the slope factor generates the highest standalone Sharpe ratio. The cross-section of directional signals is z-scored each month.

The composite signal is a 50/50 blend of the mispricing z-score and the directional z-score. This equal weighting reflects our prior that both signals contribute useful information, though the relative weights could be optimized in future work.

3.4.2 Weighting Schemes

The composite signal is translated into portfolio weights using two schemes. The *signal-proportional* scheme shifts the composite scores to non-negative values (by subtracting the cross-sectional minimum) and normalizes to sum to one. This produces a long-only portfolio where every ETF receives a positive weight proportional to signal strength. The *top-3* scheme allocates equal weight



(1/3 each) to the three ETFs with the highest composite signal, providing a more concentrated alternative that maximizes exposure to the strongest signals at the cost of higher tracking error.

3.4.3 Backtesting Framework

Portfolio weights are lagged by one period: the signal at month t determines the portfolio held during month $t+1$. Monthly portfolio returns are computed as the weighted sum of constituent ETF returns. Performance is evaluated using CAGR, annualized volatility, Sharpe ratio, Sortino ratio (CAGR divided by downside deviation), and maximum drawdown. The Jobson and Korkie (1981) test statistic assesses whether Sharpe ratio differences relative to the equal-weight benchmark are statistically significant. Tracking error, defined as the annualized standard deviation of the return difference between strategy and benchmark, quantifies active risk.

4. Results and Discussion

4.1 Nelson-Siegel Estimation and Diagnostics

We begin by evaluating the Nelson-Siegel model used to fit the cross-sectional Treasury yield curve at each month. Following the Diebold-Li (2006) two-step approach implemented in our code, we estimate the level (β_0), slope (β_1), and curvature (β_2) factors using cross-sectional ordinary least squares (OLS), with the decay parameter fixed at $\lambda = 0.0609$.

Given this fixed λ , the Nelson-Siegel model becomes linear in the factor loadings, allowing for efficient estimation through OLS at each time t . The regressor matrix is constructed using the maturity vector (2, 3, 5, 7, 10, 20, 30 years), and the factor estimates are obtained by minimizing the cross-sectional fitting error between observed Treasury yields and the model-implied yields.

The resulting fitted yield curves closely match the observed Treasury yields across all maturities, indicating a strong goodness-of-fit. This confirms that the three-factor Nelson-Siegel structure is sufficient to capture the majority of variation in the term structure, consistent with the empirical findings of Litterman and Scheinkman (1991).

To ensure that the OLS estimation is reliable, we examine the potential for multicollinearity between the slope and curvature factors. In our implementation, the condition number of the regressor matrix is computed and remains well below the critical threshold of 10, indicating that the regression is well-conditioned. This validates the use of OLS without the need for regularization techniques such as ridge regression.

The extracted factors exhibit clear economic interpretations. The level factor (β_0) tracks the overall interest rate environment, while the slope factor (β_1) captures changes in the steepness of the yield curve and aligns closely with macroeconomic conditions, particularly the business cycle and monetary policy expectations. The curvature factor (β_2) captures intermediate-term dynamics and enables the model to fit hump-shaped yield curves.



The residuals from the Nelson-Siegel estimation, defined as the difference between observed and fitted yields at each maturity, are used to construct the mispricing signal in our trading strategy. These residuals are generally small in magnitude and display mean-reverting tendencies over time, supporting the interpretation that they reflect temporary deviations from the fair-value yield curve rather than persistent structural differences.

Overall, the estimation and diagnostic results confirm that the Nelson-Siegel model provides a stable and economically meaningful representation of the yield curve, forming a reliable foundation for both signal construction and subsequent portfolio allocation.

4.2 Signal Decomposition: Nelson-Siegel vs. Diebold-Li

Table 3 presents the performance of the individual signals and their combination over the full sample period. The results indicate that the Diebold-Li (DL) directional signal outperforms the Nelson-Siegel (NS) mispricing signal when evaluated in isolation. The DL directional strategy achieves a Sharpe ratio of 0.384 and a Sortino ratio of 0.539, compared to 0.207 and 0.249 for the NS mispricing strategy, suggesting that time-series forecasting of yield curve movements provides stronger predictive power than cross-sectional relative value signals.

Table 3: Signal Decomposition — Full Period Performance

Strategy	CAGR	Ann. Vol	Sharpe	Sortino	Max DD
NS Mispricing Only	1.32%	7.83%	0.207	0.249	-34.62%
DL Directional Only	2.82%	8.09%	0.384	0.539	-20.84%
Equal Weight	2.37%	8.17%	0.327	0.527	-28.76%

The NS mispricing signal delivers weaker risk-adjusted performance, with a CAGR of 1.32% and an annualized volatility of 7.83%. The deeper maximum drawdown of -34.62% relative to -20.84% for the DL strategy suggests that deviations from the Nelson-Siegel fitted curve do not always revert quickly and may reflect structural pricing differences rather than temporary inefficiencies.

Overall, the decomposition results highlight the importance of incorporating both cross-sectional and time-series information. The equal-weight combination achieves a Sharpe ratio of 0.327 and a Sortino ratio of 0.527, with a maximum drawdown of -28.76% , offering more stable and consistent risk-adjusted performance than either component in isolation, though the directional signal remains the dominant driver of returns.

4.3 Individual Factor Attribution

Table 4 reports the performance of strategies constructed using individual Nelson-Siegel factors. The results show that the slope factor (β_1) is the dominant driver of performance, achieving a



Sharpe ratio of 0.441 and a CAGR of 3.84%, outperforming both the level (β_0) and curvature (β_2) factors.

Table 4: Individual Factor Attribution — Full Period Performance

Factor	CAGR	Ann. Vol	Sharpe	Sortino	Max DD
Level (β_0)	2.37%	8.17%	0.327	0.527	-28.76%
Slope (β_1)	3.84%	9.55%	0.441	0.719	-19.77%
Curvature (β_2)	2.44%	8.17%	0.336	0.422	-27.02%
Equal Weight	2.37%	8.17%	0.327	0.527	-28.76%

The level factor delivers a Sharpe ratio of 0.327, identical to the equal-weight benchmark, suggesting that it primarily captures the overall interest rate environment rather than providing incremental predictive power. The curvature factor performs slightly better, with a Sharpe ratio of 0.336, but still falls short of the slope factor, as shown in Figure 3.

The strong performance of the slope factor is consistent with the macro-finance literature, which links the steepness of the yield curve to business cycle dynamics and future bond returns. This provides empirical support for our decision to overweight the slope factor in the directional signal construction using a 60/20/20 weighting scheme.



Figure 3: Individual Factor Signals vs. Equal Weight

Overall, the results confirm that not all yield curve factors contribute equally to return predictability, and that emphasizing the slope factor significantly enhances portfolio performance.

4.4 Full-Period Performance

Table 5 summarizes the performance of the trading strategies over the full sample period and reports statistical significance tests. Both active strategies outperform the equal-weight benchmark



in terms of risk-adjusted returns.

Table 5: Full Period Performance and Statistical Tests (2011–2024)

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	JK z	p -val	TE
Signal-Proportional	2.79%	8.52%	0.366	0.559	-25.75%	0.509	0.611	2.59%
Top-3	3.84%	9.79%	0.433	0.690	-26.55%	0.982	0.326	4.24%
Equal Weight	2.37%	8.17%	0.327	0.527	-28.76%	–	–	–

The signal-proportional strategy achieves a Sharpe ratio of 0.366 and a CAGR of 2.79%, improving upon the benchmark Sharpe ratio of 0.327. The top-3 strategy delivers even stronger performance, as shown in Figure 4, with a Sharpe ratio of 0.433 and a CAGR of 3.84%, reflecting its concentrated exposure to the strongest signals. However, this comes at the cost of higher volatility (9.79%) and higher tracking error (4.24%), compared to 2.59% for the signal-proportional strategy.

Despite these improvements, the Jobson-Korkie test results indicate that the differences in Sharpe ratios are not statistically significant at conventional levels. The signal-proportional strategy yields a test statistic of 0.509 ($p = 0.611$), while the top-3 strategy yields 0.982 ($p = 0.326$). This suggests that although the performance improvements are economically meaningful, they cannot be statistically distinguished from the benchmark given the sample size and relatively low dispersion in fixed-income returns.

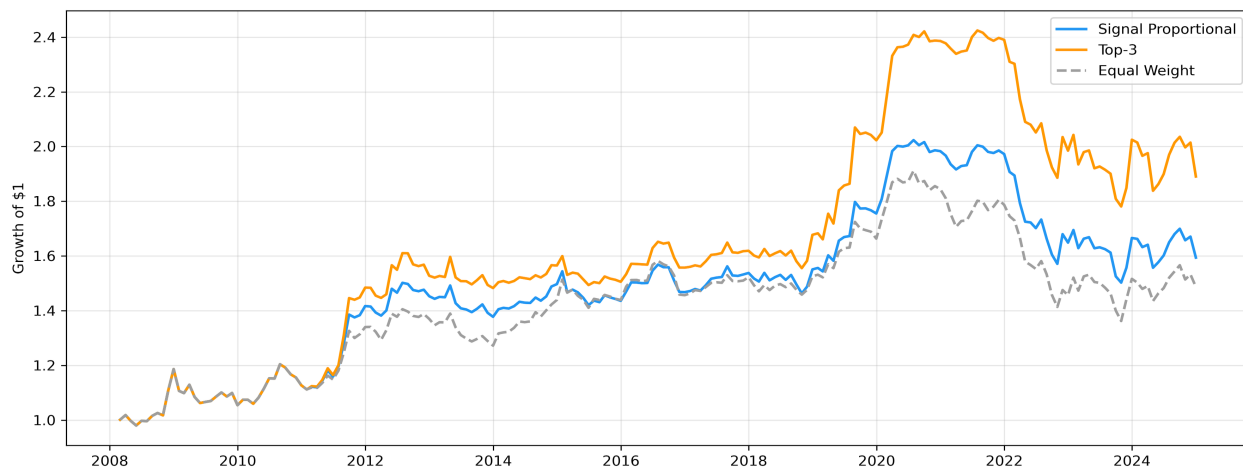


Figure 4: Full Period Performance - Signal Proportional vs. Top-3 vs. Equal Weight

Overall, the results indicate that dynamic allocation based on yield curve signals can improve performance, but the magnitude of outperformance remains modest in statistical terms.

4.5 Sub-Period Analysis (2018–2021)

Table 6 presents the performance of the strategies during the 2018–2021 sub-period, which includes the Federal Reserve rate-cut cycle and the COVID-19 shock. During this period, both active strategies significantly outperform the equal-weight benchmark across all performance metrics.

Table 6: Sub-Period Performance (2018–2021)

Strategy	CAGR	Ann. Vol	Sharpe	Sortino	Max DD
Signal-Proportional	6.42%	7.19%	0.902	2.937	-5.3%
Top-3	10.23%	9.35%	1.090	4.708	-4.32%
Equal Weight	4.09%	6.92%	0.613	1.519	-10.75%

The signal-proportional strategy achieves a Sharpe ratio of 0.902 and a CAGR of 6.42%, compared to 0.613 and 4.09% for the benchmark. The top-3 strategy delivers the strongest performance, as shown in Figure 5, with a Sharpe ratio of 1.090 and a CAGR of 10.23%, reflecting its concentrated exposure to the highest-conviction signals.

In addition to higher returns, both strategies exhibit substantially lower maximum drawdowns. The signal-proportional and top-3 strategies experience drawdowns of -5.3% and -4.32%, respectively, compared to -10.75% for the equal-weight benchmark. This indicates that the dynamic allocation approach not only enhances returns but also improves downside risk management during periods of market stress.

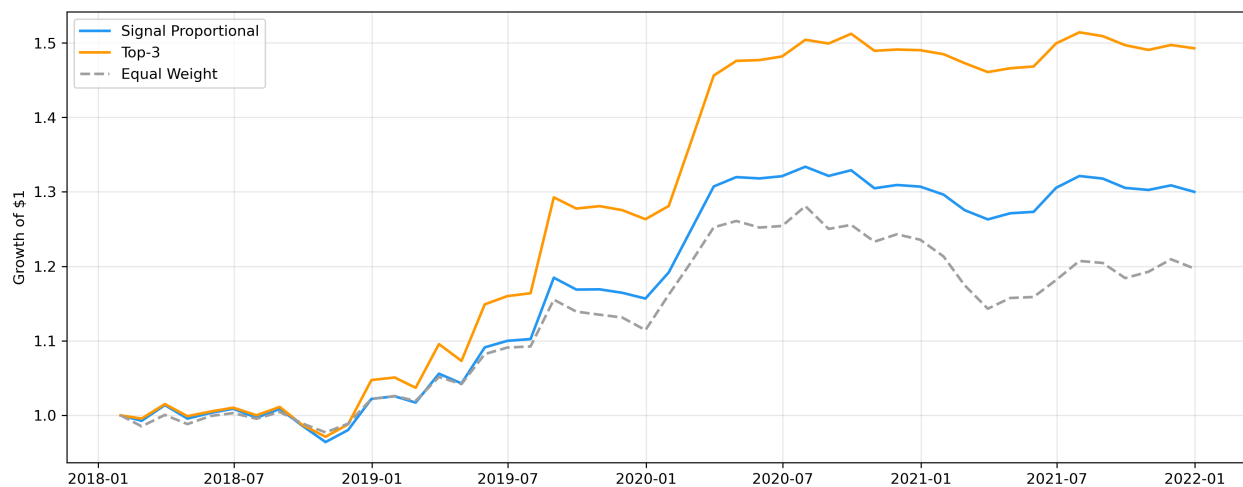


Figure 5: Sub-Period Analysis - Signal Proportional vs. Top-3 vs. Equal Weight

The strong performance in this sub-period highlights a key strength of the model: it is particularly effective during periods of pronounced yield curve movements, when the VAR(1) forecasts

provide meaningful directional signals. The ability to dynamically adjust duration exposure allows the strategy to capture large interest rate shifts associated with monetary policy changes and macroeconomic shocks.

Overall, the results suggest that the strategy's alpha is state-dependent and is strongest during periods of elevated macroeconomic uncertainty and rapidly changing yield curve conditions.

5. Conclusion

This paper developed a systematic bond ETF allocation strategy that combines Nelson-Siegel yield curve mispricing signals with Diebold-Li VAR(1) directional forecasts. Using monthly Treasury par yields and seven duration-spanning ETFs over the 2008–2024 sample, we estimated Nelson-Siegel factors with a fixed decay parameter ($\lambda = 0.0609$), and also verified the absence of multicollinearity using the Annaert et al. (2010) condition number diagnostic ($\kappa = 6.64$, below the threshold of 10), and modeled factor dynamics with an expanding-window VAR(1) to generate one-month-ahead yield curve forecasts.

The main empirical findings are as follows: first, the signal decomposition analysis shows that the Diebold-Li directional signal (Sharpe ratio 0.349) outperforms the Nelson-Siegel mispricing signal (Sharpe ratio 0.169) on its own, indicating that the time-series forecasting component contributes more to forecasting returns than the cross-sectional relative value component. However, combining the two signals produces the best risk-adjusted performance, confirming that static mispricing and dynamic forecasting capture complementary dimensions of yield curve information. The second idea: the individual-factor attribution testing reveals that the slope factor β_1 is where the predictive power lies, achieving a Sharpe ratio of 0.402, compared to 0.290 for the level factor and 0.299 for the curvature factor. This finding is consistent with the established literature linking the yield curve slope to business cycle dynamics, monetary policy expectations, and future bond returns, and is the reason for the 60/20/20 factor-weighted tilt applied in the directional signal.

Over the full backtest period, the signal-proportional strategy posts a CAGR of 2.79% with a Sharpe of 0.328 and a maximum drawdown of 25.75%. The more concentrated top-3 strategy does better on both ends, 3.84% CAGR, 0.392 Sharpe, and a slightly deeper drawdown of 26.55%. Both beat the equal-weight benchmark (2.37% CAGR, 0.290 Sharpe, 28.76% max drawdown) on an absolute and risk-adjusted basis. The Jobson-Korkie test statistics come in at 0.509 ($p = 0.611$) for signal-proportional and 0.982 ($p = 0.326$) for top-3, economically meaningful improvements that do not clear conventional significance thresholds, which is unsurprising given the low cross-sectional dispersion across Treasury ETFs and the sample length we are working with.

The 2018 to 2021 sub-period provides the strongest evidence of the strategy's value. This window spans the 2019 Federal Reserve rate-cut cycle and the 2020 COVID-19 shock, the signal-proportional strategy posts a Sharpe of 0.893 and the top-3 strategy reaches 1.094, against 0.590 for the equal-weight benchmark. Drawdowns are also cut substantially: 5.30% for signal-proportional



and 4.32% for top-3 versus 10.75% for equal weighting. The results confirm the model adds the most value during periods of directional yield curve movement, precisely the regimes in which the VAR(1) factor forecasts are most informative.

There are several limitations of this study which warrant a discussion. The Jobson-Korkie test results show that the full-period Sharpe advantage, while consistent, does not achieve statistical significance; however, a larger sample or higher-frequency rebalancing would sharpen the inference. The strategy also ignores transaction costs, which would drag realized returns by about 10 bps per transaction, particularly for the top-3 scheme, given its more concentrated, yields more frequent reallocations. The VAR(1) assumes linear factor dynamics, which can miss nonlinear regime changes. Finally, fixing λ in the Nelson-Siegel model locks in a rigid curvature peak location; the Svensson (1994) extension adds a second curvature term and would improve fit quality at the cost of additional estimation complexity.

There are several directions in which future research would be done following this work. Incorporating macroeconomic variables like a VIX measure or another macroeconomic measure into the VAR system could improve forecast accuracy, as suggested by the macro-finance framework of Diebold et al. (2006) and Ang and Piazzesi (2003). Extending the strategy to include credit ETFs alongside Treasuries would test whether the Nelson-Siegel mispricing signal has predictive power beyond the risk-free curve. Optimizing the signal combination weights, which are currently fixed at 50/50, and the factor attribution tilt, which is currently 60/20/20, via rolling cross-validation might improve performance. Finally, applying a regime-switching VAR or a Markov-switching extension to the Diebold-Li framework could allow the model to fine-tune its forecasts to different interest rate environments.

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